Syllabus: Introduction to Operational Amplifiers, Op-Amp Input Modes and Parameters, Op-Amp with Negative Feedbacks, Comparator (Zero-Level Detection and Nonzero-Level Detection), Summing Amplifier with Unity Gain, The Op-Amp Integrator, The Op-Amp Differentiator. (Text 3: 12.1-12.4, 13.1 – 13.3)

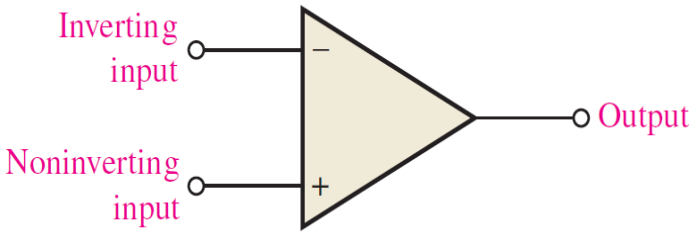
Module - 4 : Operational Amplifiers

Introduction to Operational Amplifiers

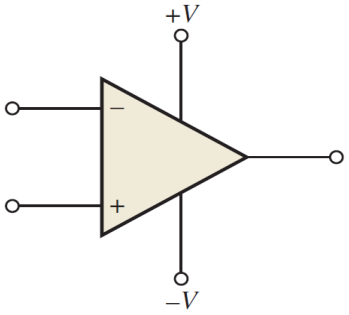
- Early operational amplifiers (op-amps) were used primarily to perform mathematical operations such as addition, subtraction, integration, and differentiation.
- These early devices were constructed with vacuum tubes and worked with high voltages.
- Today's op-amps are linear integrated circuits (ICs) that use relatively low dc supply voltages and are reliable and inexpensive.

Symbol:

- The standard operational amplifier (op-amp) symbol is shown in figure below:



- It has two input terminals, the inverting (-) input and the non inverting (+) input, and one output terminal.
- Most op-amps operate with two dc supply voltages, one positive and the other negative, as shown in figure below:

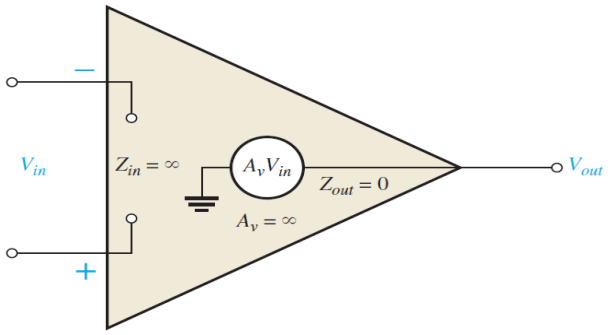


The Ideal Op-Amp:

The ideal op-amp has,

- Infinite Voltage Gain
- Infinite Bandwidth
- Infinite Input Impedance
- Zero Output Impedance
- Infinite Slew Rate
- Infinite Common Mode Rejection Ratio (CMRR)

- Op-amp characteristics are illustrated in figure below:
- The input voltage, V_{in}, appears between the two input terminals, and the output voltage is $A_v V_{in}$, as indicated by the internal voltage source symbol.

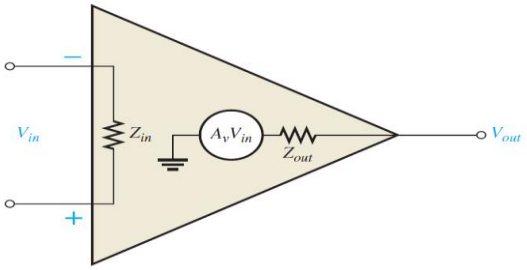


The Practical Op-Amp:

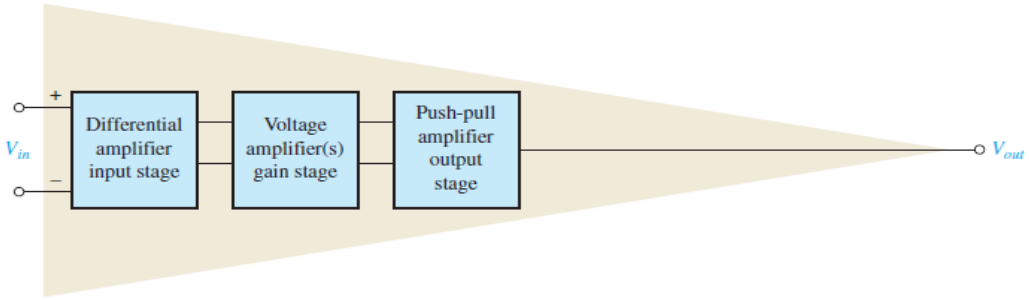
- Although integrated circuit (IC) op-amps approach parameter values that can be treated as ideal in many cases, the ideal device can never be made.
- IC op-amp has practical limitations.
- Op-amps have both voltage and current limitations.
- Peak to- peak output voltage, is usually limited to slightly less than the two supply voltages.
- Output current is also limited by internal restrictions component such as power dissipation and ratings.

Characteristics of a practical op-amp are

- Very High Voltage Gain
- Very High Input Impedance
- Very Low Output Impedance
- High Bandwidth



Internal Block Diagram of an Op-Amp:



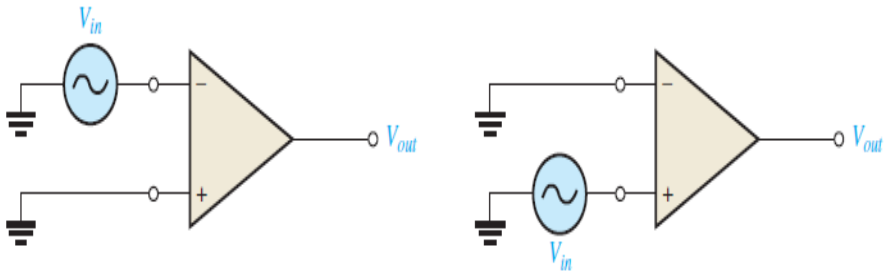
- A typical op-amp is made up of three types of amplifier circuits: a differential amplifier, a voltage amplifier, and a push-pull amplifier.
- The differential amplifier is the input stage for the op-amp. It provides amplification of the difference voltage between the two inputs.
- The second stage is usually a class A amplifier that provides additional gain. Some op-amps may have more than one voltage amplifier stage.
- A push-pull class B amplifier is typically used for the output stage.
- The term differential comes from the amplifier's ability to amplify the difference of two input signals applied to its inputs.
- Only the difference in the two signals is amplified; if there is no difference, the output is zero.
- The differential amplifier exhibits two modes of operation based on the type of input signals.

- These modes are –Differential and Common.

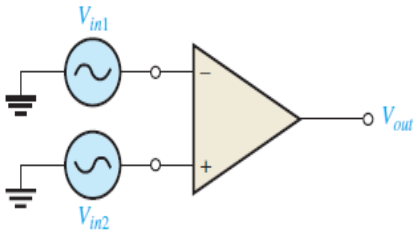
Op-amp Input Modes and Parameters:

Differential Mode:

- In the differential mode, either one signal is applied to an input with the other input grounded or two opposite-polarity signals are applied to the inputs.
- When an op-amp is operated in the single-ended differential mode, one input is grounded and a signal voltage is applied to the other input, as shown in figure below:

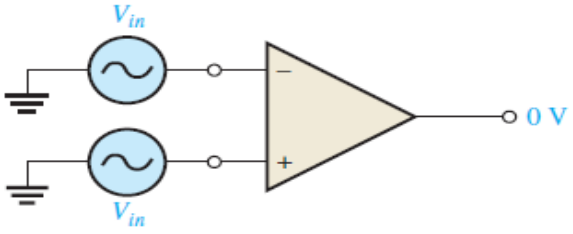


- In the case where the signal voltage is applied to the inverting input, an inverted, amplified signal voltage appears at the output.
- In the case where the signal is applied to the non- inverting input with the inverting input grounded, a non-inverted, amplified signal voltage appears at the output.
- In the double-ended differential mode, two opposite-polarity (out-of-phase) signals are applied to the inputs. The amplified difference between the two inputs appears on the output.
- Equivalently, the double-ended differential mode can be represented by a single source connected between the two inputs, as shown in figure.



Common Mode:

- In the common mode, two signal voltages of the same phase, frequency, and amplitude are applied to the two inputs, as shown in figure:



- When equal input signals are applied to both inputs, they tend to cancel, resulting in a zero output voltage.
- This action is called common-mode rejection. Its importance lies in the situation where an unwanted signal appears commonly on both op-amp inputs.

- Common-mode rejection means that this unwanted signal will not appear on the output and distort the desired signal.

Op-Amp Parameters:

Common-Mode Rejection Ratio (CMRR):

- Unwanted signals (noise) appearing with the same polarity on both input lines are essentially cancelled by the op-amp and do not appear on the output.
- The measure of an amplifier's ability to reject common-mode signals is a parameter called the CMRR (common-mode rejection ratio).
- Ideally, an op-amp provides a very high gain for differential-mode signals and zero gain for common-mode signals.
- Practical op-amps, however, do exhibit a very small common-mode gain (usually much less than 1), while providing a high open-loop differential voltage gain (usually several thousand).
- The higher the open-loop gain with respect to the common-mode gain, the better the performance of the op-amp in terms of rejection of common-mode signals.
- This suggests that a good measure of the op-amp's performance in rejecting unwanted common-mode signals is the ratio of the open-loop differential voltage gain, A_{ol}, to the common-mode gain, A_{cm}. This ratio is the common-mode rejection ratio, CMRR.

$$\mathbf{CMRR} = \frac{A_{ol}}{A_{cm}}$$

- The higher the CMRR, the better. A very high value of CMRR means that the open-loop gain, A_{ol}, is high and the common-mode gain, A_{cm}, is low.
- The CMRR is often expressed in decibels (dB) as

$$\mathbf{CMRR} = 20 \log \left(\frac{A_{ol}}{A_{cm}} \right)$$

- The open-loop voltage gain, A_{ol}, of an internal voltage gain of the device and op-amp is the represents the ratio of output voltage to input voltage when there are no external components.
- The open-loop voltage gain is set entirely by the internal design. Open-loop voltage gain can range up to 200,000 (106 dB) and is not a well-controlled parameter.

Problems:

1. A certain op-amp has an open-loop differential voltage gain of 100,000 and a common-mode gain of 0.2. Determine the CMRR and express it in decibels.
   ANS:500,000 . In decibels =114dB

2. Determine the CMRR and express it in dB for an op -amp with an open-loop differential voltage gain of 85,000 and a common-mode gain of 0.25.
   ANS: 3,40,000 . In decibels =110.6dB

Maximum Output Voltage Swing (Vo(p-p)):

- With no input signal, the output of an op amp is ideally 0 V. This is called the *quiescent output* voltage.
- When an input signal is applied, the peak-to-peak output signals have the ideal limits of ±VCC.
- In practice, however, this ideal can be approached but never reached.
- Vo(p-p) varies with the load connected to the op-amp and increases directly with load resistance.
- Some op-amps do not use both positive and negative supply voltages. In this case, the op-amp

output is designed to operate between ground and a full scale output that is near (or at) the positive supply voltage.

- Op-amps that operate on a single supply use the terminology VOH and VOL to specify the maximum and minimum output voltage.

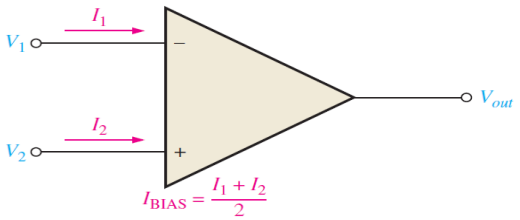
Input Offset Voltage (V_{OS}):

- The ideal op-amp produces zero volts out for zero volts in.
- In a practical op-amp, however, a small dc voltage, V_{OUT}(error), appears at the output when no differential input voltage is applied.
- Its primary cause is a slight mismatch of the base emitter voltages of the differential amplifier input stage of an op-amp.
- The input offset voltage, V_{OS}, is the differential dc voltage required between the inputs to force the output to zero volts.
- Typical values of input offset voltage are in the range of 2 mV or less. In the ideal case, it is 0 V.
- The input offset voltage drift is a parameter related V_{OS} that specifies how much change occurs in the input offset voltage for each degree change in temperature.
- Typical values range anywhere from about 5µV per degree Celsius to about per 50µV degree Celsius.

Input Bias Current (I_{BIAS}):

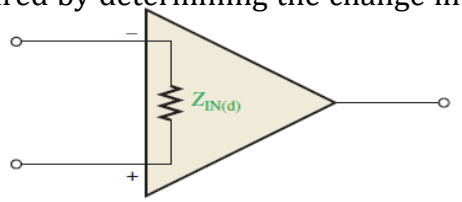
- The input bias current is the dc current required by the inputs of the amplifier to properly operate the first stage.
- By definition, the input bias current is the average of both input currents and is calculated as follows:

$$I_{BIAS} = \frac{I_1 + I_2}{2}$$

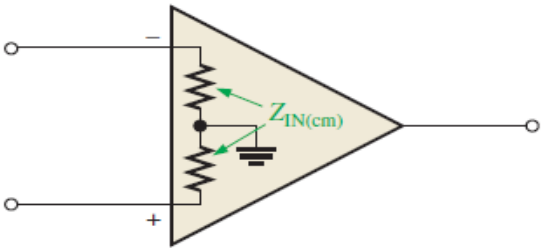


Input Impedance:

- Two basic ways of specifying the input impedance of an op-amp are the differential and the common mode.
- The differential input impedance is the total resistance between the inverting and the non inverting inputs:
- Differential impedance is measured by determining the change in bias current for a given change in differential input voltage.



- The common-mode input impedance is the resistance between each input and ground and is measured by determining the change in bias current for a given change in common-mode input voltage

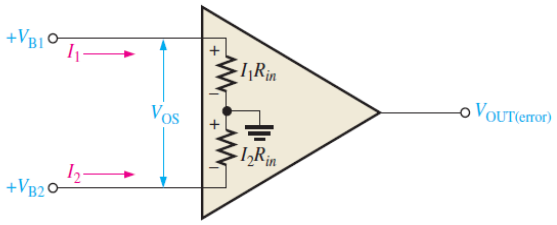


Input Offset Current (I_{OS}):

- Ideally, the two input bias currents are equal, and thus their difference is zero.
- In a practical op-amp, however, the bias currents are not exactly equal.
- The input bias offset current, I_{OS}, is the difference of the currents, expressed as an absolute value.

$$I_{OS} = |I_1 - I_2|$$

- Actual magnitudes of offset current are usually at least an order of magnitude (ten times) less than the bias current.
- In many applications, the offset current can be neglected.
- However, high-gain, high-input impedance amplifiers should have difference in as little I_{OS} as currents through possible because the large input resistances develops a substantial offset voltage, as shown in Figure



- The offset voltage developed by the input offset current is

$$V_{OUT(error)} = A_v I_{OS} R_{in}$$

- The error created by I_{OS} is amplified by the gain Av of the op-amp and appears in the output as

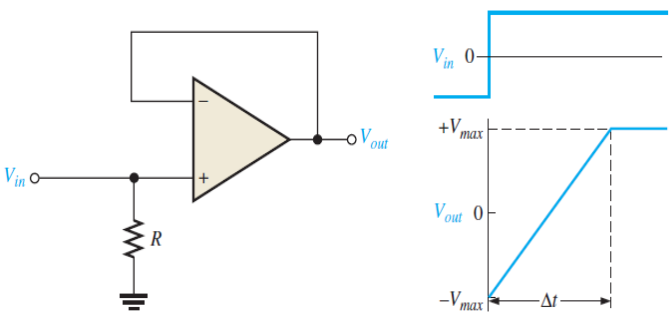
$$V_{OS} = I_1 R_{in} - I_2 R_{in} = (I_1 - I_2) R_{in}$$

$$V_{OS} = I_{OS} R_{in}$$

- A change in offset current with temperature affects the error voltage.

Slew Rate:

- The maximum rate of change of the output voltage in response to a step input voltage is the slew rate of an op-amp.
- The slew rate is dependent upon the high-frequency response of the amplifier stages within the op-amp.
- Slew rate is measured with an op-amp connected as shown in Figure



- A pulse is applied to the input and the resulting ideal output voltage is indicated in figure.
- The width of the input pulse must be sufficient to allow the output to "slew" from its lower limit to its upper limit.

- A certain time interval, Δt, is required for the output voltage to go from its lower limit –Vmax to its upper limit +Vmax once the input step is applied.
- The slew rate is expressed as

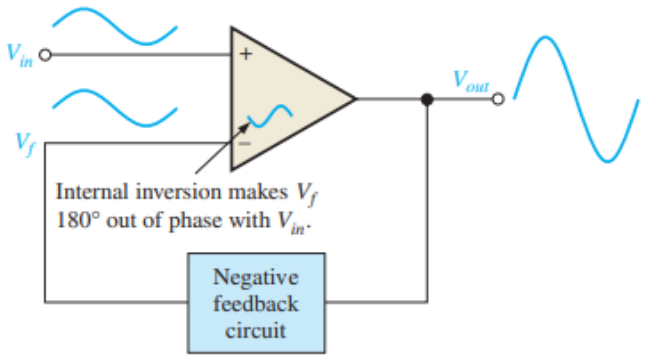
$$\textbf{Slew rate} = \frac{\Delta V_{out}}{\Delta t}$$

where $\Delta V_{out} = +V_{max} - (-V_{max})$. The unit of slew rate is volts per microsecond (V/μs).

Output Impedance:
The output impedance is the resistance viewed from the output terminal of the op-amp.

Negative Feedback

Negative feedback is the process whereby a portion of the output voltage of an amplifier is returned to the input with a phase angle that opposes (or subtracts from) the input signal.



- The inverting input effectively makes the feedback signal 180° out of phase with the input signal.
- The usefulness of an op-amp operated without negative feedback is generally limited to comparator applications. With negative feedback, op-amp can function as a linear amplifier. Following table
- summarizes the general effects of negative feedback on op-amp performance.

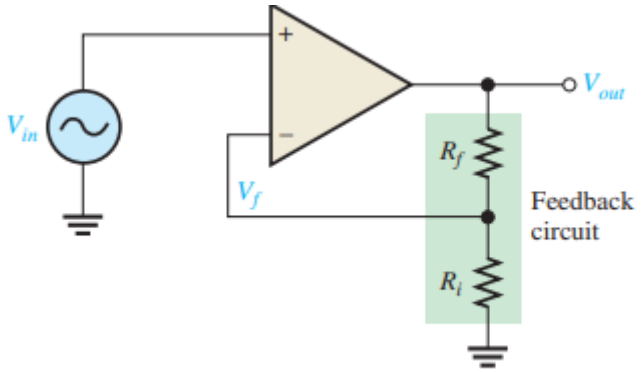
	VOLTAGE GAIN	INPUT Z	PUTPUT Z	BANDWIDTH
With negative feedback	A_{ol} is too high for linear amplifier applications	Relatively high	Relatively low	Relatively narrow (because the gain is so high)
With negative feedback	A_{cl} is set to desired value by the feedback circuit	Can be increased or reduced to a desired value depending on type of circuit	Can be reduced to a desired value	Significantly wider

An op-amp can be connected using negative feedback to stabilize the gain and increase frequency response. Negative feedback takes a portion of the output and applies it back out of phase with the input, creating an effective reduction in gain. This closed loop gain is usually much less than the open-loop gain and independent of it.

Closed-Loop Voltage Gain, A_{cl} :

The closed-loop voltage gain is the voltage gain of an op-amp with external feedback. The amplifier configuration consists of the op-amp and an external negative feedback circuit that connects the output to the inverting input.

Noninverting Amplifier:



The input signal is applied to the noninverting (+) input. The output is applied back to the inverting input through the feedback circuit (closed loop) formed by the input resistor R_i and the feedback resistor R_f. Resistors R_i and R_f form a voltage-divider circuit, which reduces V_{out} and connects the reduced voltage V_f to the inverting input. The feedback voltage is expressed as

$$V_f = \left(\frac{R_i}{R_i + R_f}\right)V_{out}$$

The difference of the input voltage, V_{in}, and the feedback voltage, V_f, is the differential input to the op-amp. This differential voltage is amplified by the open-loop voltage gain of the op-amp (A_{ol}) and produces an output voltage expressed as

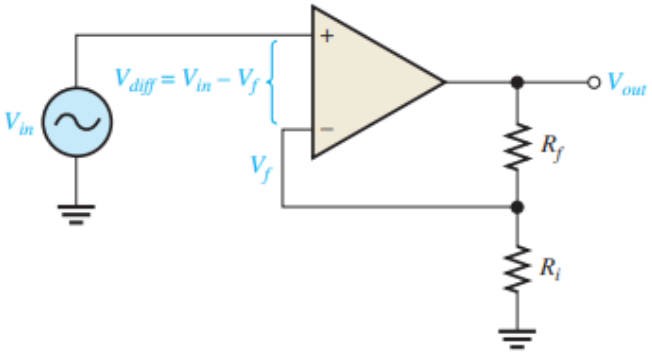
$$V_{out} = A_{ol}(V_{in} - V_f)$$

The attenuation, B, of the feedback circuit is

$$B = \frac{R_i}{R_i + R_f}$$

Substituting BV_{out} for V_f in the V_{out} equation,

$$V_{out} = A_{ol}(V_{in} - BV_{out})$$



Then applying basic algebra,

$$V_{out} = A_{ol}V_{in} - A_{ol}BV_{out}$$
$$V_{out} + A_{ol}BV_{out} = A_{ol}V_{in}$$
$$V_{out}(1 + A_{ol}B) = A_{ol}V_{in}$$

Since the overall voltage gain of the amplifier in Figure is V_{out}/V_{in}, it can be expressed as

$$\frac{V_{out}}{V_{in}} = \frac{A_{ol}}{1 + A_{ol}B}$$

The product $A_{ol}B$ is typically much greater than 1, so the equation simplifies to

$$\frac{V_{out}}{V_{in}} \cong \frac{A_{ol}}{A_{ol}B} = \frac{1}{B}$$

The closed-loop gain of the noninverting (NI) amplifier is the reciprocal of the attenuation (B) of the feedback circuit (voltage-divider).

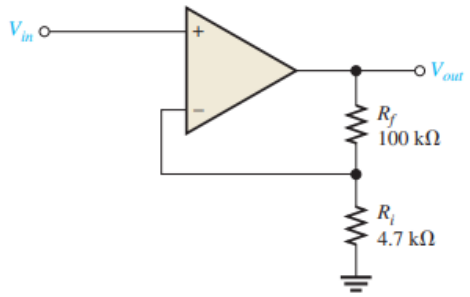
$$A_{cl(\text{NI})} = \frac{V_{out}}{V_{in}} \cong \frac{1}{B} = \frac{R_i + R_f}{R_i}$$

Therefore,

$$\boldsymbol{A_{cl(\text{NI})} = 1 + \frac{R_f}{R_i}}$$

Problem:

1. Determine the closed-loop voltage gain of the amplifier in Figure



Solution:

This is a noninverting op-amp configuration. Therefore, the closed-loop voltage gain is

$$\boldsymbol{A_{cl(\text{NI})} = 1 + \frac{R_f}{R_i}}$$
$$= \mathbf{1} + \frac{\mathbf{100k\Omega}}{\mathbf{4.7k\Omega}} = \underline{22.3}$$

2. If R_f in above Figure is increased to 150kΩ determine the closed-loop gain

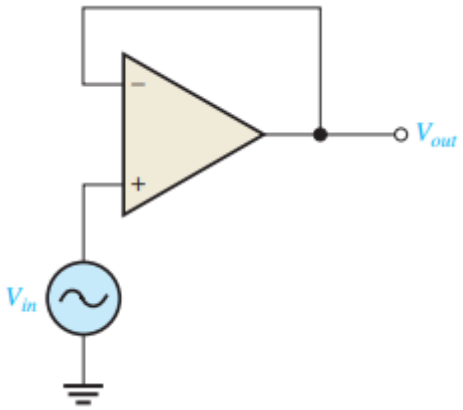
Solution:

$A_{cl(\text{NI})} =$ <u>32.91</u>

Voltage-Follower:

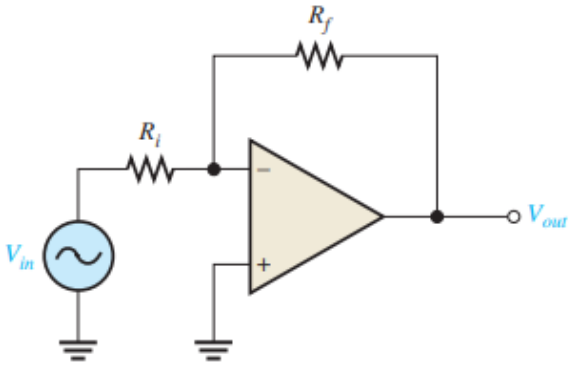
The voltage-follower configuration is a special case of the noninverting amplifier where all of the output voltage is fed back to the inverting input by a straight connection. The straight feedback

connection has a voltage gain of 1 (which means there is no gain). Therefore , $\mathbf{A_{cl(VF)}= 1}$

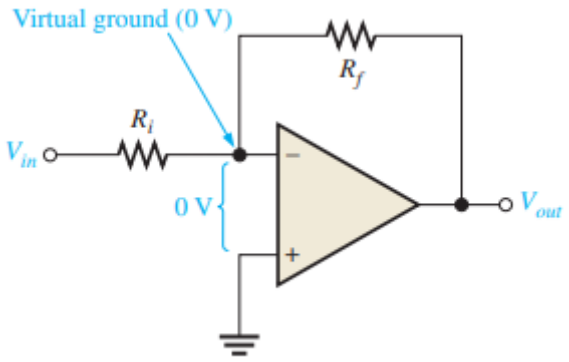


The most important features of the voltage-follower configuration are its very high input impedance and its very low output impedance. These features make it a nearly ideal buffer amplifier for interfacing high-impedance sources and low-impedance loads.

Inverting Amplifier:



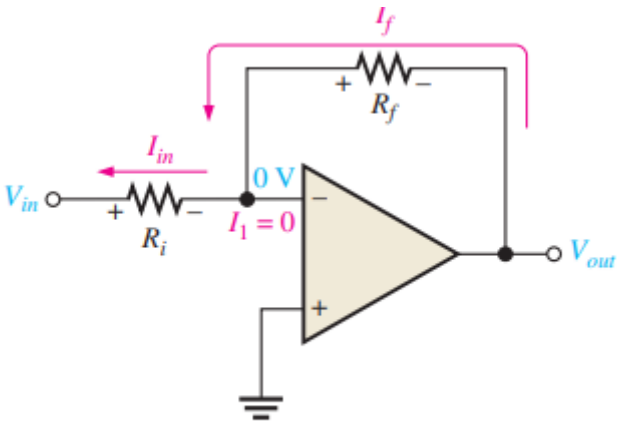
The input signal is applied through a series input resistor R_i to the inverting input. Also, the output is fed back through R_f to the same input. The noninverting (+) input is grounded.
An infinite input impedance implies zero current at the inverting input. If there is zero current through the input terminal, then there must be no voltage drop between the inverting and noninverting inputs. This means that the voltage at the inverting input is zero because the noninverting (+) input is grounded. This zero voltage at the inverting input terminal is referred to as *virtual ground* which is shown in below figure a.



(a) Virtual ground

Since there is no current at the inverting input, the current through R_i and the current through R_f are equal, as shown in fig.b

$$\mathbf{I_{in} = I_f}$$



(b) $I_{in} = I_f$ and current at the inverting input (I_1) is 0.

The voltage across R_i equals V_{in} because the resistor is connected to virtual ground at the inverting input of the op-amp. Therefore,

$$I_{in} = \frac{V_{in}}{R_i}$$

Also, the voltage across R_f equals $-V_{out}$ because of virtual ground, and therefore,

$$I_f = \frac{-V_{out}}{R_f}$$

Since $I_f = I_{in}$,

$$\frac{-V_{out}}{R_f} = \frac{V_{in}}{R_i}$$

Rearranging the terms,

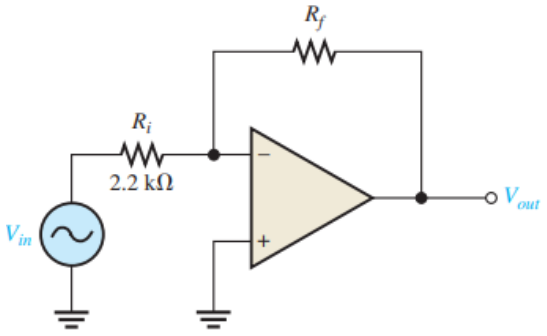
$$\frac{V_{out}}{V_{in}} = -\frac{R_f}{R_i}$$

Of course, V_{out}/V_{in} is the overall gain of the inverting (I) amplifier.

$$A_{cl(I)} = -\frac{R_f}{R_i}$$

Problem:

1. Given the op-amp configuration in Figure, determine the value of R_f required to produce a closed-loop voltage gain of -100.



Solution:

Knowing that $R_i = 2.2\ \text{k}\Omega$ and the absolute value of the closed-loop gain is $|A_{cl(\text{I})}| = 100$, calculate R_f as follows:

$$|A_{cl(\text{I})}| = \frac{R_f}{R_i}$$

$$R_f = |A_{cl(\text{I})}|R_i = (100)(2.2\ \text{k}\Omega) = \mathbf{220\ k\Omega}$$

Comparator:

A *comparator* is a specialized op-amp circuit that compares two input voltages and produces an output that is always at either one of two states, indicating the greater or less than relationship between the inputs.

Comparators provide very fast switching times, and many have additional capabilities (such as fast propagation delay or internal reference voltages) to optimize the comparison function.

In general, comparators cannot be used as op-amps, but op-amps can be used as comparators in noncritical applications. Because an op-amp without negative feedback is essentially a comparator.

Zero-Level Detection:

One application of an op-amp used as a comparator is to determine when an input voltage exceeds a certain level.

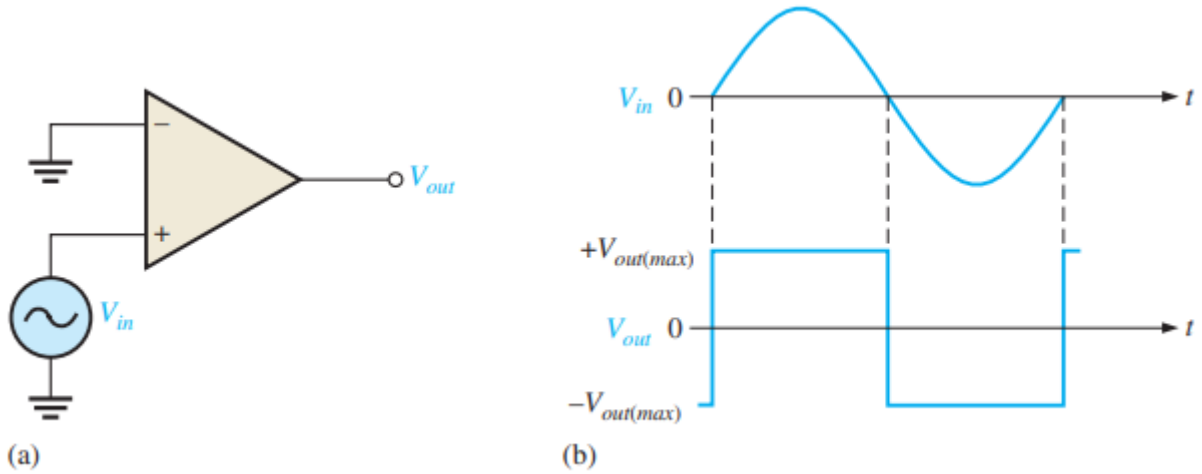


Figure (a) shows a zero-level detector. Here inverting input is grounded to produce a zero level and that the input signal voltage is applied to the noninverting input. Because of the high open-loop voltage gain, a very small difference voltage between the two inputs drives the amplifier into saturation, causing the output voltage to go to its limit. Figure (b) shows the result of a sinusoidal input voltage applied to the noninverting input of the zero-level detector. When the sine wave is positive, the output is at its maximum positive level. When the sine wave crosses 0, the amplifier is driven to its opposite state and the output goes to its maximum negative level, as shown.

The zerolevel detector can be used as a squaring circuit to produce a square wave from a sine wave.

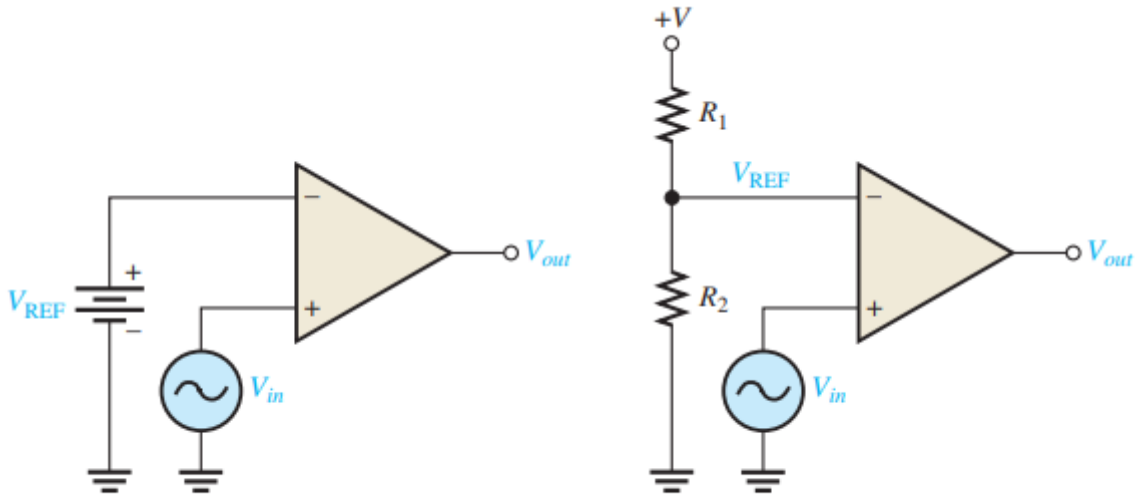
Nonzero-Level Detection:

The zero-level detector can be modified to detect positive and negative voltages by connecting a fixed reference voltage source to the inverting input, as shown in Figure (a) and a more practical arrangement is shown in Figure (b) using a voltage divider to set the reference voltage, V_{REF}, as

follows:

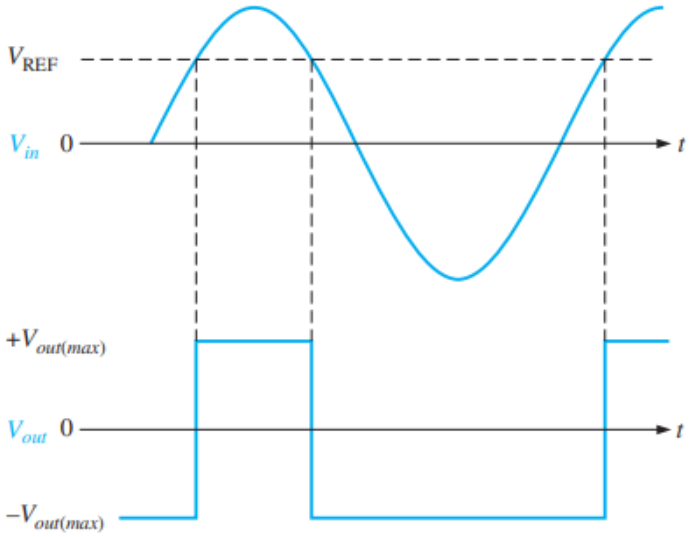
$$V_{REF} = \frac{R_2}{R_1 + R_2}(+V)$$

where +V is the positive op-amp dc supply voltage.



(a) Battery reference (b) Voltage-divider reference

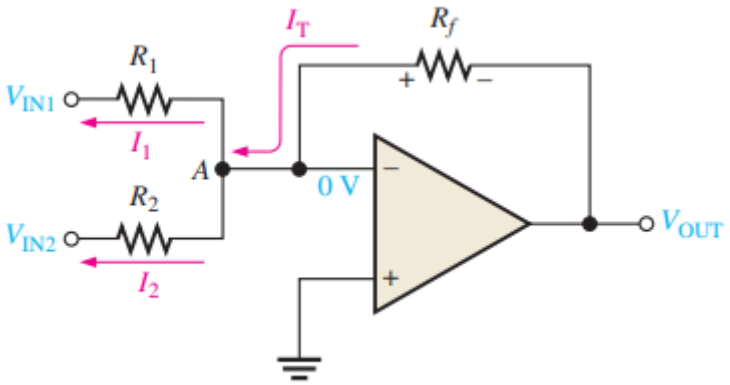
As long as V_{in} is less than V_{REF}, the output remains at the maximum negative level. When the input voltage exceeds the reference voltage, the output goes to its maximum positive voltage which is shown in below waveform.



Summing Amplifier with Unity Gain:

A summing amplifier has two or more inputs, and its output voltage is proportional to the negative of the algebraic sum of its input voltages.

Two voltages, V_{IN1} and V_{IN2}, are applied to the inputs and produce currents I_1 and I_2, as shown.



Using the concept of infinite input impedance and virtual ground, the inverting input of the op-amp is approximately 0 V and has no current through it. This means that the total current I_T, which goes through R_f divides into I_1 and I_2 at summing point A.

$$I_T = I_1 + I_2$$

Since $V_{OUT}=-I_TR_f$ the following steps apply:

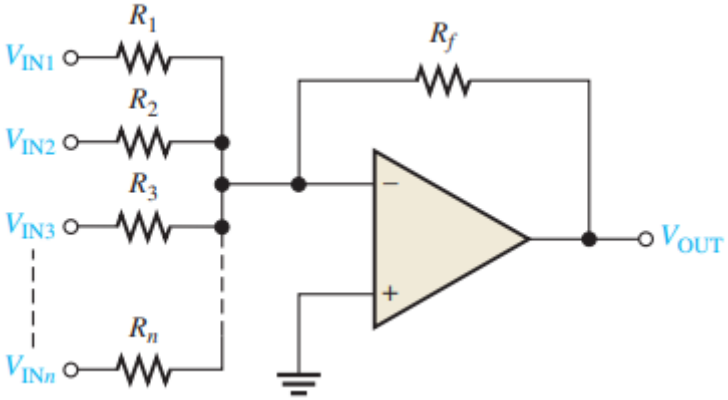
$$V_{OUT} = -(I_1 + I_2)R_f = -\left(\frac{V_{IN1}}{R_1} + \frac{V_{IN2}}{R_2}\right)R_f$$

If all three of the resistors are equal ($R_1 = R_2 = R_f = R$), then

$$V_{OUT} = -\left(\frac{V_{IN1}}{R} + \frac{V_{IN2}}{R}\right)R = -(V_{IN1} + V_{IN2})$$

The previous equation shows that the output voltage has the same magnitude as the sum of the two input voltages but with a negative sign, indicating inversion.
For a unity-gain summing amplifier with n inputs, as shown, all resistors are equal in value. Therefore,

$$V_{OUT} = -(V_{IN1} + V_{IN2} + V_{IN3} + \cdots + V_{INn})$$



Summing Amplifier with Gain Greater Than Unity

When R_f is larger than the input resistors, the amplifier has a gain of R_f /R, where R is the value of each equal-value input resistor. The general expression for the output is

$$V_{OUT} = -\frac{R_f}{R}(V_{IN1} + V_{IN2} + \cdots + V_{INn})$$

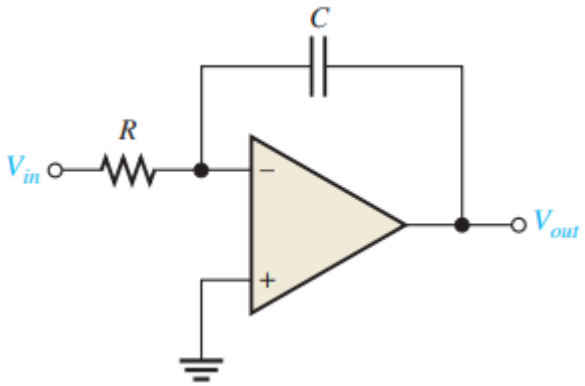
Note: Please refer textbook for problems related to summing amplifier.

Integrator and differentiators:

An op-amp *integrator* simulates mathematical integration, which is basically a summing process that determines the total area under the curve of a function.

An op-amp differentiator simulates mathematical differentiation, which is a process of determining the instantaneous rate of change of a function.

The Op-Amp Integrator:



The feedback element is a capacitor that forms an RC circuit with the input resistor. The charge Q on a capacitor is proportional to the charging current (I_C) and the time (t).

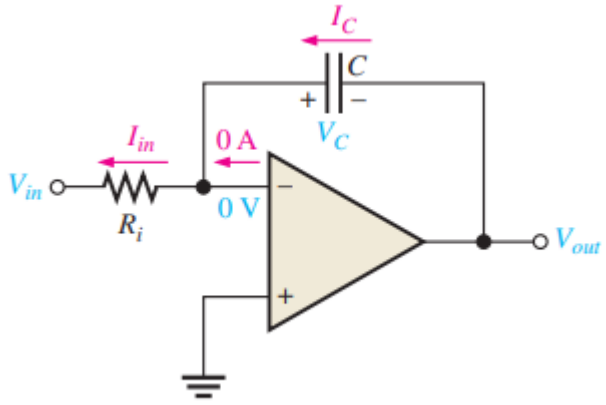
$$Q = I_C t$$

Also, in terms of the voltage, the charge on a capacitor is

$$Q = CV_C$$

From these two relationships, the capacitor voltage can be expressed as

$$V_C = \left(\frac{I_C}{C}\right)t$$



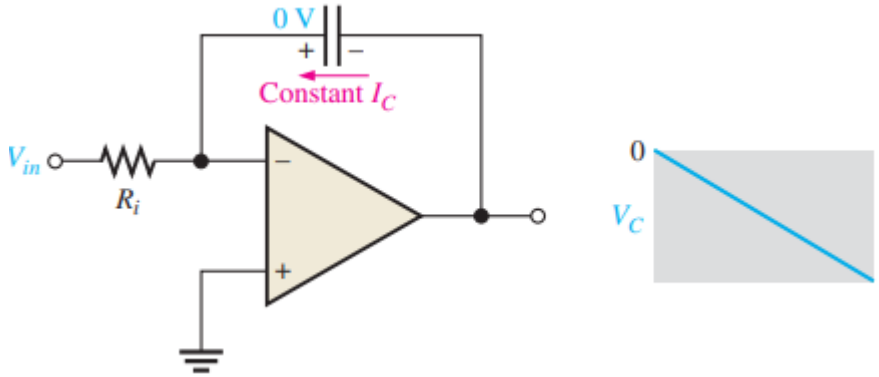
In the above fig. the inverting input of the op-amp is at virtual ground (0 V), so the voltage across R_i equals V_{in}. Therefore, the input current is

$$I_{in} = \frac{V_{in}}{R_i}$$

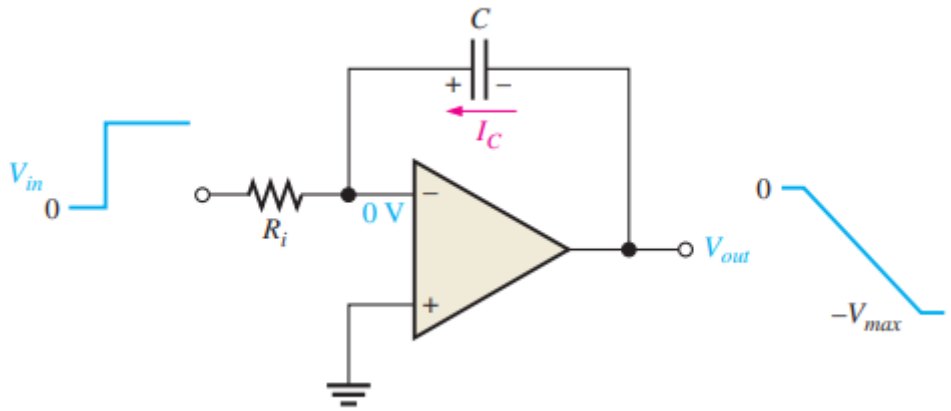
If V_{in} is a constant voltage, then I_{in} is also a constant because the inverting input always remains at 0 V, keeping a constant voltage across R_i. Because of the very high input impedance of the op-amp, there is negligible current at the inverting input.

$I_C = I_{in}$

Since I_{in} is constant, so is I_C. The constant I_C charges the capacitor linearly and produces a linear voltage across C. The positive side of the capacitor is held at 0 V by the virtual ground of the op-amp. The voltage on the negative side of the capacitor, which is the op-amp output voltage, decreases linearly from zero as the capacitor charges, This voltage, V_C, is called a negative ramp and is the consequence of a constant positive input.



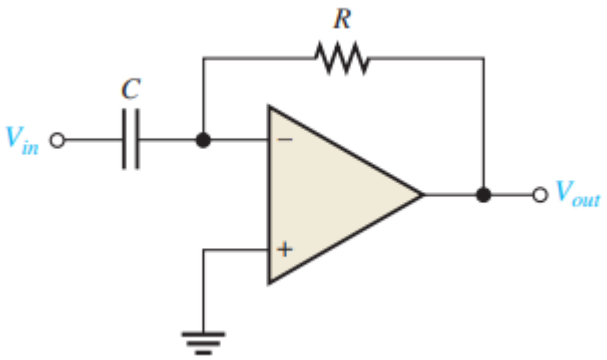
V_{out} is the same as the voltage on the negative side of the capacitor. When a constant positive input voltage in the form of a step or pulse (a pulse has a constant amplitude when high) is applied, the output ramp decreases negatively until the op-amp saturates at its maximum negative level.



The rate at which the capacitor charges, and therefore the slope of the output ramp, is set by the ratio I_C/C,. Since $IC = V_{in}/R_i$, the rate of change or slope of the integrator's output voltage is

$$\frac{\Delta V_{out}}{\Delta t} = -\frac{V_{in}}{R_i C}$$

The Op-Amp Differentiator:



The capacitor isthe input element, and the resistor is the feedback element. A differentiator produces an output that is proportional to the rate of change of the input voltage.

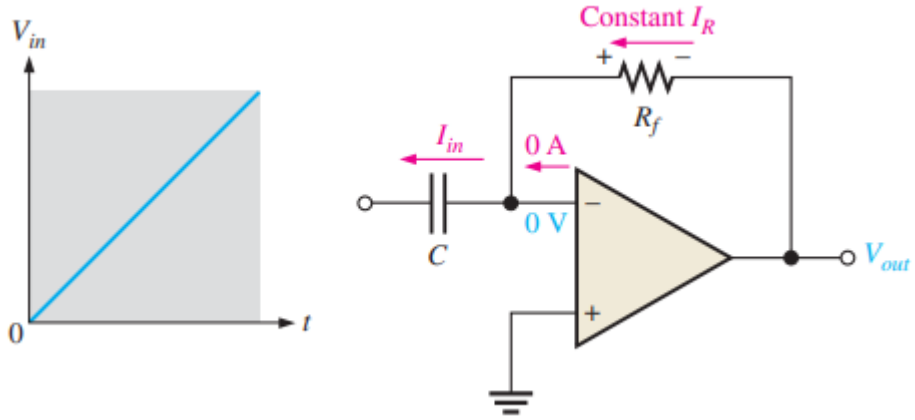
Working:

Apply a positive-going ramp voltage to the input.

In this case, $I_C = I_{in}$ and the voltage across the capacitor is equal to V_{in} at all times ($V_C = V_{in}$) because of virtual ground on the inverting input.

From the basic formula, $V_C = (I_C/C)t$, the capacitor current is

$$I_C = \left(\frac{V_C}{t}\right)C$$



Since the current at the inverting input is negligible, $I_R = I_C$. Both currents are constant because the slope of the capacitor voltage (V_C/t) is constant. The output voltage is also constant and equal to the voltage across R_f because one side of the feedback resistor is always 0 V (virtual ground)

$$V_{out} = I_R R_f = I_C R_f$$

$$\boldsymbol{V_{out} = -\left(\frac{V_C}{t}\right)R_f C}$$

The output is negative when the input is a positive-going ramp and positive when the input is a negative-going ramp. During the positive slope of the input, the capacitor is charging from the input source and the constant current through the feedback resistor is in the direction shown. During the negative slope of the input, the current is in the opposite direction because the capacitor is discharging.

